

# Vendor Module – Revised Database Structure

## Generic Multi-Category Product Listing + Separate School Approval/Mapping Table

School Uniform & Supplies Marketplace – Revised Design (June 2026)

Update applied per your feedback: (1) Vendor adds a product into a single, category-generic Product table so a vendor can sell uniforms, shoes, ties, belts, and also stationery items such as copies, pens, and books — all using one flexible structure. (2) The School-side approval and class/gender mapping is kept in a completely separate table (School–Product Mapping & Approval), so “vendor adds product” and “school approves & maps product” remain two distinct steps/tables.

### 1. Product Category Table (NEW – enables "sell anything")

A separate Category master table allows the platform to support any product type — clothing, footwear, accessories, or stationery — without changing the schema. Each category can define which extra attributes apply (e.g. size\_chart\_type, requires\_size).

| Field Name      | Data Type    | Description                                                                          | Validation / Notes                         |
|-----------------|--------------|--------------------------------------------------------------------------------------|--------------------------------------------|
| category_id     | UUID         | Unique category identifier                                                           | Auto-generated, Primary Key                |
| category_name   | VARCHAR(100) | e.g. Shirt, Trousers, Skirt, Tie, Belt, Shoes, Notebook, Pen, Textbook, Bag          | Required, unique                           |
| category_type   | ENUM         | uniform / footwear / accessory / stationery / other                                  | Required – groups categories for filtering |
| requires_size   | BOOLEAN      | Whether this category needs size selection (e.g. shirts, shoes = true; pens = false) | Default: true                              |
| size_chart_type | ENUM         | age_based / numeric / standard(S-XL) / shoe_size / not_applicable                    | Optional, only if requires_size = true     |
| is_active       | BOOLEAN      | Category enabled for new listings                                                    | Default: true                              |
| created_at      | DATETIME     | Record creation date                                                                 | Auto-set                                   |

### 2. Vendor Table (unchanged)

Stores vendor registration, KYC, GST/PAN, bank settlement, and commission details. Referenced by Product and Order tables via vendor\_id.

| Field Name                  | Data Type        | Description                    | Validation / Notes          |
|-----------------------------|------------------|--------------------------------|-----------------------------|
| vendor_id                   | UUID             | Unique vendor identifier       | Auto-generated, Primary Key |
| business_name               | VARCHAR(150)     | Shop / brand name              | Required, unique            |
| owner_name                  | VARCHAR(100)     | Owner full name                | Required                    |
| email                       | VARCHAR(150)     | Login / contact email          | Unique, required            |
| phone                       | VARCHAR(15)      | Mobile number                  | 10 digits, required         |
| gstn / pan_number           | VARCHAR(20)/(15) | GST & PAN identifiers          | Format-validated            |
| bank_account_no / ifsc_code | VARCHAR(25)/(15) | Settlement bank details        | Encrypted storage           |
| commission_rate             | DECIMAL(5,2)     | Platform commission %          | Range 0–100                 |
| status                      | ENUM             | pending / approved / suspended | Set by Admin                |
| created_at                  | DATETIME         | Registration date              | Auto-set                    |

### 3. Product Table (REVISED – Vendor Listing, Any Category)

This single table is used by a vendor to list ANY item — uniform shirts/skirts, ties, belts, shoes, bags, or stationery (notebooks, pens, textbooks, etc.). The category\_id field determines what type of item it is. There is NO school/class information here — that lives entirely in the separate Mapping & Approval table (Section 4). This table only represents “what the vendor is selling.”

| Field Name             | Data Type        | Description                                                                               | Validation / Notes                                          |
|------------------------|------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| product_id             | UUID             | Unique product identifier                                                                 | Auto-generated, Primary Key                                 |
| vendor_id              | FK → vendor_id   | Vendor who listed the product                                                             | Required (Foreign Key)                                      |
| category_id            | FK → category_id | Type of item: shirt, tie, shoes, pen, notebook, etc.                                      | Required (Foreign Key) – NEW                                |
| product_name           | VARCHAR(200)     | Product title (e.g. "School Tie - Striped Red/White")                                     | Required                                                    |
| gender_type            | ENUM             | boys / girls / unisex                                                                     | Required for uniform items; default 'unisex' for stationery |
| sizes_available        | JSON             | Array of sizes/variants: [S,M,L,XL] or [28,30,32] for shoes, [10,12,14,16] for bags, etc. | Optional, but recommended for clothing/bags                 |
| unit_type              | VARCHAR(30)      | piece / pair / set / pack — e.g. shoes sold as 'pair', pens as 'pack of 5'                | Default: 'piece'                                            |
| price                  | DECIMAL(10,2)    | MRP per unit                                                                              | Positive number                                             |
| discount_price         | DECIMAL(10,2)    | Discounted selling price                                                                  | Less than price, optional                                   |
| gst_rate               | DECIMAL(5,2)     | GST percentage (e.g. 5, 12, 18)                                                           | Valid GST slab                                              |
| stock_qty              | INT              | Total available stock                                                                     | Non-negative                                                |
| images                 | JSON             | Array of image URLs                                                                       | Min 1 image required                                        |
| description            | TEXT             | Product description                                                                       | Optional                                                    |
| fabric_type / material | VARCHAR(80)      | Cotton/Polyester for clothing; Leather/Cloth for shoes/bags; Paper for stationery         | Optional, but useful for filtering                          |
| approval_status        | ENUM             | pending / approved / rejected                                                             | Set by ADMIN only (platform-level QC)                       |
| is_active              | BOOLEAN          | Listed / unlisted by vendor                                                               | Default: true                                               |
| created_at             | DATETIME         | Listing date                                                                              | Auto-set                                                    |

## 4. School–Product Mapping & Approval Table (SEPARATE TABLE)

This is a completely separate table from Product, exactly as you described: the Vendor adds the product (Section 3) independently of any school. Then, separately, a School owner reviews admin-approved products and decides which ones to approve and map for their own school, by class and gender. One product can be mapped to MANY schools, each with its own approval decision — this is a many-to-many relationship table.

| Field Name      | Data Type        | Description                                   | Validation / Notes                                                     |
|-----------------|------------------|-----------------------------------------------|------------------------------------------------------------------------|
| mapping_id      | UUID             | Unique mapping record                         | Auto-generated, Primary Key                                            |
| school_id       | FK → school_id   | Which school this approval/mapping belongs to | Required (Foreign Key)                                                 |
| product_id      | FK → product_id  | Which vendor product is being considered      | Required (Foreign Key); must have product.approval_status = 'approved' |
| class_name      | VARCHAR(20)      | Specific class (e.g. Class 5) or 'ALL'        | Required for product filter; not used for non-uniform items            |
| gender_type     | ENUM             | boys / girls / unisex                         | Optional filter, inherited or overridden from product                  |
| is_mandatory    | BOOLEAN          | Whether item is a compulsory uniform item     | Default: optional/stationery item                                      |
| academic_year   | VARCHAR(10)      | e.g. 2025-26                                  | Year-wise mapping                                                      |
| school_decision | ENUM             | pending / approved / rejected                 | NEW – School owner's own decision, independent of product              |
| reviewed_by     | FK → school user | School staff/owner who approved/rejected      | NEW – audit trail                                                      |
| reviewed_at     | DATETIME         | When school made the decision                 | NEW – set on approval/rejection                                        |
| created_at      | DATETIME         | Mapping record creation date                  | Auto-set                                                               |

Visibility Rule: a product becomes visible to a parent ONLY if ALL of the following are true — product.approval\_status = 'approved' (Admin), vendor.status = 'approved', product.is\_active = true, product.stock\_qty > 0, AND a matching School–Product Mapping row exists with school\_decision = 'approved' for that student's school\_id + class\_name + gender.

## 5. Revised Vendor → Admin → School Flow

### Step 1 – Vendor Adds Product (Product table)

Vendor logs in, selects a category\_id (clothing, footwear, accessory, or stationery — anything from shirts to pens), and fills product\_name, gender\_type, sizes\_available, unit\_type, price, gst\_rate, stock\_qty, images, material. Saved with approval\_status = 'pending'. NO school is involved at this stage.

### Step 2 – Admin Reviews Product (Product table)

Admin checks product quality/compliance and sets approval\_status to 'approved' or 'rejected'. This is a platform-wide check, independent of any specific school.

### Step 3 – School Browses Approved Catalog

School owner logs into School Portal and views the list of all Admin-approved products across all categories (uniforms, ties, shoes, stationery, etc.).

### Step 4 – School Creates Mapping & Approval (separate Mapping table)

For each product the school wants for its students, the school creates a NEW row in the School–Product Mapping & Approval table, choosing class\_name, gender\_type, is\_mandatory, academic\_year, and setting school\_decision = 'approved' (or 'rejected' to skip it).

### Step 5 – Product Becomes Visible to Parents

Only products with a matching, school-approved Mapping row appear in that school's parent catalog, filtered further by class and gender.

### Step 6 – Orders, Inventory & Settlement

Parents order; vendor manages stock\_qty and fulfils via the Order table as before. Sales reports and commission tracking work the same regardless of category (clothing or stationery).

Summary: **Vendor adds product (any category) → Admin approves product → School separately reviews & creates its own approval/mapping record (class + gender + academic year) → Product visible to matching parents → Order & fulfilment.**

## 6. Updated Total Database Tables

With the new Category table and the clearly separated School–Product Mapping & Approval table, the platform now uses 9 core tables.

| # | Table Name                        | Purpose                                                                                                         | Key Relationships                                           |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1 | Admin                             | Platform admin/sub-admin accounts and roles                                                                     | Referenced by School.onboarded_by                           |
| 2 | Vendor                            | Vendor registration, KYC, GST, bank & com                                                                       | Referenced by Product, Order                                |
| 3 | Category                          | Master list of product types (uniform, footwear, accessories, stationery, etc.)                                 | Referenced by Product (category_id) ← NEW                   |
| 4 | Product                           | Vendor-listed items of ANY category (clothes, shoes, ties, belts, bags, stationery (copies, pens, books), etc.) | school_id, vendor_id → Category; referenced by Student, Map |
| 5 | School                            | School registration, address, classes_offered, state                                                            | onboarded_by → Admin; referenced by Student, Map            |
| 6 | School–Product Mapping & Approval | SEPARATE table: school's own approval + class selection mapping for product                                     | school_id, product_id → Product                             |
| 7 | Parent                            | Guardian login accounts (email/phone OTP)                                                                       | Referenced by Student, Order                                |
| 8 | Student Profile                   | Child profile linked to parent + school + class + gender                                                        | parent_id → Parent; school_id → School; referenced by Order |
| 9 | Order                             | Order/transaction records with items, payment, status                                                           | student_id, school_id, vendor_id → respective tables        |

**Total core tables (revised): 9** — Admin, Vendor, Category (NEW), Product (now generic for any item type), School, School–Product Mapping & Approval (separate, with its own school\_decision field), Parent, Student Profile, Order.

Note: This is a revised design based on the original "School Uniform Marketplace – Platform Design & Technical Specification" (v1.0, June 2026), updated to (1) separate vendor product listing from school approval/mapping into two distinct tables, and (2) generalize the Product table via a new Category table so vendors can sell any item type — uniforms, shoes, ties, belts, bags, stationery (copies, pens, books), etc.